

FinTrust Digital Bank - Week 4 Project Documentation

Data Science Track: FinTrust Final Predictive Risk Model

Project: FinTrust Financial Intelligence & Digital Banking Support Solution

Submitted by: Emmanuella Esi Arhin

A. Application of the Week 1-3 Foundation

What I originally planned to accomplish. In Week 1 I defined the predictive problem: can the information known about a transaction predict whether it is marked for risk review? In Week 4 the plan was to test, validate, refine and finalize the model, compare my findings with the Data Analytics work, and document everything for the final FinTrust solution.

What Weeks 2 and 3 delivered.

- Week 2: a prepared dataset, nine EDA charts, eight engineered features and a balanced Logistic Regression baseline (12 features, ROC-AUC 0.668).
- Week 3: a chronological 60/20/20 split, five engineered features, four tuned model families, bootstrap intervals and a candidate Logistic Regression (test ROC-AUC 0.672, PR-AUC 0.321, precision 0.360, recall 0.366 at a 20% review budget), with the Week 3 modelling dataset `FinTrust_Modelling_Dataset_v2.csv`.

How Week 4 builds on it. Week 4 continues directly from the Week 3 dataset. I loaded `FinTrust_Modelling_Dataset_v2.csv` and checked that my preparation reproduces it exactly on all 12,000 rows (IDs and order, split labels, target, features, amounts and history columns). I also rebuilt the Week 3 candidate from the raw files and matched its submitted numbers exactly (cut-off NGN 142,058; confusion matrix 1,637 / 301 / 293 / 169). The final dataset is the Week 3 dataset, refined on purpose, not a new dataset.

Changes to my approach and why.

- **A second evaluation protocol.** The Week 3 test period had already been viewed, so I added a nested rolling-origin evaluation (five expanding windows, 7,200 predictions on later data) as the main evidence. The hold-out test is used for confirmation only.
- **A new model-selection rule.** Week 3's rule (within 0.01 of the best) was borderline for Logistic Regression, so I replaced it with a rule written in advance (see Part B).
- **Calibration and a safe scoring function.** Both were Week 3 limitations that I closed this week.

B. Use of the Approved Project Resources

Resource	How it was used	Limitation encountered
<code>FinTrust_Modelling_Dataset_v2.csv</code> (Week 3)	Starting point; reproduced exactly, then refined	Customer attributes carry no signal; three-month window
<code>FinTrust_Transaction_Data</code>	Re-derivation of features	96 missing

Resource	How it was used	Limitation encountered
and Customer_Data	inside each training window; sensitivity tests and the DA comparison	Device_Type/Location values filled with "Unknown"
FinTrust_Data_Dictionary	Confirmed Customer_Name is not for modelling	Covers customer fields only
Week 1-3 documentation and notebooks	Baselines, decisions and open issues carried forward	None
Data Analytics intern's SQL script and 3-page Power BI dashboard (screenshots)	Reconciled to the raw data and compared with predictive tests	No SQL result tables were included
Project Brief, Roadmap, Week 4 assignment	Scope, requirements and responsible-use rules	Knowledge Base not used (GenAI resource)

No ML Engineering or GenAI output was received, and no integration with those tracks is claimed.

C. Tools Used

Python 3.12, Jupyter in VS Code, pandas, NumPy, scikit-learn 1.8 (pipelines, Logistic Regression, Decision Tree, Random Forest, HistGradientBoosting, GridSearchCV, TimeSeriesSplit, permutation importance), SciPy (chi-square tests), Matplotlib and Seaborn, joblib, and Git/GitHub.

D. Practical Outputs Produced

- **Final notebook** FinTrust_Week4_DS_Final_Model.ipynb (executed, covering Parts A-F, testing, integration and limitations).
- **Final modelling dataset** FinTrust_Modelling_Dataset_Final.csv: the Week 3 dataset, refined.
- **Final model, calibrator and model card** (final_model.joblib, final_calibrator.joblib, final_model_card.json) with an input and output contract.
- **Validated raw-record scoring function** with 15 functional tests.
- **Five key charts** in reports/figures; every other table and chart is in the notebook.

E. Week 4 Development Work

Part A - Final feature review

I tested twenty feature variants (the incumbent plus 19 changes). Each was a Logistic Regression fitted inside five expanding windows and scored on the next block, compared with the incumbent on the same 500 bootstrap resamples. The adoption rules were written before the results were read.

- **Four signals earned their place.** Removing transaction type costs 0.055 ROC-AUC, night-time 0.028, the high-amount flag 0.020 and is_international 0.008. All intervals are entirely below zero.
- **The amount effect is a step.** A scan picks NGN 160,000 in all five windows. The Week 3 features (log-amount and the 90th-percentile flag) score 0.006 ROC-AUC lower, with an interval that excludes zero.
- **Nothing added helped reliably.** Log-amount, hour, weekend, channel and device, interactions, demographics, past-only history, the look-ahead counts, Account_Status and the Week 3 recency column log_hours_since_prev (-0.0004 [-0.0020, +0.0010]) all have intervals that include zero.
- **Transaction_Status is the one exception.** It adds +0.006, but it may not be known when a transaction is scored, so it stays out and its cost is documented.
- **A judgement call, disclosed.** Collapsing the six transaction types into "Transfer/Cash vs other" is statistically non-inferior (-0.0003). I kept the six types because collapsing would discard business-readable distinctions.
- **Final features (4 inputs):** Transaction_Type, high_amount ($\geq$ NGN 160,000), is_international, is_night (00:00-05:59). This is down from 12 features in Week 2 and 5 in Week 3.
- **Changes to the Week 3 dataset:** the cut-off moved from NGN 142,058 to NGN 160,000 (197 rows change); log_amount, amount_vs_cust_avg, prior_txn_count and log_hours_since_prev were removed; Transaction_Status was added as a reference column only.

Part B - Final model evaluation

Eight models plus a signal-count scorecard were compared, with hyper-parameters tuned by time-series cross-validation inside every training window.

Model	Nested ROC-AUC (n = 7,200)	Hold-out precision / recall at 20% budget	Hold-out ROC-AUC
Gradient Boosting (extended features)	0.676 [0.660, 0.690]	0.360 / 0.374	0.688
Week 4 Logistic Regression (final)	0.674 [0.658, 0.688]	0.357 / 0.368	0.679
Random Forest	0.672	0.357 / 0.377	0.678
Decision Tree	0.672	0.337 / 0.411 (flags 23.5% of transactions, above a strict 20% budget)	0.673
Week 3 candidate	0.668	0.360 / 0.366	0.672

Model	Nested ROC-AUC (n = 7,200)	Hold-out precision / recall at 20% budget	Hold-out ROC-AUC
Week 2 recipe	0.667	0.327 / 0.331	0.677

- **Selection rule (written in advance):** a more complex model replaces Logistic Regression only if a paired bootstrap shows it better on both ROC-AUC and PR-AUC with the interval above zero. No challenger qualified. The extended Gradient Boosting that led in Week 3 (0.691) comes out at +0.001 [-0.005, +0.007] under nested evaluation, so its Week 3 lead did not reproduce.
- **Final model:** Logistic Regression, chosen for simplicity and explainability, not accuracy. It beats the Week 3 candidate by 0.006 ROC-AUC (interval excludes zero) in the nested evaluation. On the hold-out alone the difference (+0.007 [-0.003, +0.017]) cannot be separated.
- **Hold-out results (final model, 20% budget):** accuracy 0.751, precision 0.357, recall 0.368, F1 0.362, ROC-AUC 0.679, PR-AUC 0.325, lift 1.84x. Accuracy is below the 0.808 of "always No" and is not a meaningful metric here.
- **Review budget (per 1,000 transactions):** a 10% budget gives precision 0.41; 20% gives 0.35 (71 of 200 reviewed are truly flagged, 122 flagged transactions missed); 30% gives 0.31; 50% gives 0.28. The 20% default is an assumption, not a requirement.
- **Calibration:** raw scores averaged 0.47 against a flag rate of 0.19. A Platt layer fitted on the validation period cut the calibration error (ECE) from 0.281 to 0.021 and the Brier score from 0.226 to 0.145. Ranking is unchanged.

Part C - Error analysis (hold-out, 20% budget)

- **Confusion matrix:** TN 1,632, FP 306, FN 292, TP 170.
- **The model counts risk signals.** With two or more signals a transaction is always flagged, with none never, with one 4.8% of the time.
- **False negatives are quiet transactions.** All 292 have one signal or none, with a median amount near NGN 9k. The actual flag rate is 8.4% with no signal and 21.4% with one, so these labels cannot be explained from the available fields.
- **False positives look like true positives.** With two signals the true flag rate is only about 36%, so 306 of 476 flagged transactions (64%) are false alarms.
- **Strong and weak segments.** All international transactions and nearly all amounts of NGN 160k or more are flagged (precision 32% and 41%). Recall is only 9-13% for Airtime/Data, Bill Payment, Card Purchase and Deposit.
- **A data limit, not a model limit.** A flexible model given every field reaches ROC-AUC 0.681 against 0.679 for the final model.
- **Subgroups.** Selection shares are 0.18-0.23 across gender, segment, account type, income band and city, with attribute-level tests giving p of 0.44 or higher.

Part D - Final model interpretation

- **Odds ratios (signal present vs absent, 95% bootstrap interval):** international 3.34 [2.42, 4.26], high amount 2.87 [2.36, 3.54], night-time 2.28 [1.97, 2.57]. Transfer (1.63) and

Cash Withdrawal (1.40) raise risk; Card Purchase (0.72), Bill Payment (0.71), Deposit (0.67) and Airtime/Data (0.83) lower it.

- **Importance:** high amount first, then transaction type, night-time and international, consistently across Logistic Regression, Random Forest and Gradient Boosting. International has the largest odds ratio but ranks fourth because only about 4% of transactions are international.
- **Strengths:** fully explainable, stable over time (ROC-AUC 0.674 +/- 0.010), no look-ahead, identifiers or customer attributes.
- **Weaknesses:** a modest ceiling, many false alarms, and scores that tie because few distinct profiles exist.
- **Assumptions:** Q1 2026 patterns persist; the cut-off, night window and budget are fixed inputs; the label is a consistent synthetic rule. These are associations with a synthetic label, not causes.

Part E - Data Analytics comparison

I received the Data Analytics intern's SQL script and Power BI dashboard. All 74 values printed on the dashboard reproduce from the raw data, which I checked before using any of them.

- **Agree:** transaction type is a core driver (removing it costs 0.055 ROC-AUC). The dashboard's finding that international transactions are flagged about twice as often (36.88% vs 18.88%) holds in all six transaction types (gaps of 14-23 percentage points), and an explicit international-by-type interaction adds nothing, so it is not a type effect in disguise.
- **Agree (stable):** demand is steady at about 133 transactions a day, and the flag rate is steady across months (18.5%-20.3%, $p = 0.10$). Segments are near-identical on value per customer, and demographics add no predictive value (ROC-AUC 0.501 alone). The top-10 customers hold only 2.6% of value, and the model does not depend on customer identity.
- **Differ:** the dashboard shows ATM terminals drawing the most risk reviews (21.64%), but channel and device add no predictive gain once the four signals are known.
- **Signal real, not used:** failure is a real predictive signal (+0.006) but is only known after processing.
- **Not on any DA page:** night-time (flag rate about 28% vs 17%) and the NGN 160,000 amount step (about 18% vs 40%). I have five suggestions for the dashboard (these two views, a device caption, the page-3 gauge maximum and a city-gap caption); they are offered, not yet shared.
- **What changed:** the comparison prompted two extra tests (the international interaction and flag rate by month). It did not change the final features or model.

Part F - Final model documentation

- **Final model:** Logistic Regression (balanced weights, $C = 1$) plus a Platt calibration layer, on four inputs.

- **Operating points:** a 20% review budget (default) and an F1-optimal high-recall mode that flags about 61% of transactions.
- **Model card:** documents intended and not-intended use, features and exclusions, thresholds, calibration, metrics, limitations and the input and output contract.
- **Scoring function:** accepts raw transaction fields, rejects invalid records (unknown type, missing, negative or non-numeric amount, bad timestamp, bad Yes/No) and routes them to manual check. It reproduces the evaluated model's scores on all 2,400 test records.
- **Responsible use:** decision support only; a human reviews every flag; no customer attributes are used.

F. Testing / Evaluation of the Work

Work completed	Test / evaluation	Finding	Action / improvement	Result
Week 3 candidate	Re-evaluated under the nested protocol with paired bootstrap	The 90th-percentile cut-off sits on the edge of a plateau; log-amount adds nothing	Learned cut-off NGN 160,000; removed log-amount	ROC-AUC +0.006 [+0.002, +0.010]; equal or better in 4 of 5 windows
Raw scores	Reliability check, Brier score, ECE	Scores overstate risk (0.47 vs 0.19)	Platt layer on the validation period	ECE 0.281 to 0.021; ranking unchanged
Model interface	15 functional tests (valid, boundary, bad input, empty, missing column)	Passed 7 of 15; invalid records were scored silently	Validated scoring function with reject-and-route	15 of 15 pass; scores identical to the evaluated model
Selection rule	Paired bootstrap against Logistic Regression	Week 3 rule was borderline	Burden-of-proof rule	No challenger qualifies
Leakage and memorisation	Shuffled-label test; grouped vs random CV	Mean ROC-AUC 0.484 on shuffled labels; 0.676 vs 0.677	None needed	No leakage or customer memorization
Stability	Five rolling windows; threshold applied to test	ROC-AUC 0.674 +/- 0.010 (range 0.659-0.686); flags	None needed	Stable

Work completed	Test / evaluation	Finding	Action improvement /	Result
		20.4% of validation vs 19.8% of test		

Required chain (Final output, Test, Finding, Refinement, Re-test, Final result): the Week 3 candidate was re-tested, found to depend on a data-driven cut-off and a redundant feature, refined to four inputs, re-tested on the same windows and resamples, and came out slightly better and simpler. The model reload test, raw-scoring equivalence and subgroup checks also passed.

G. Important Decisions

Decision	Reason	Evidence	Effect on the work
Continue from the Week 3 dataset	The assignment asks to build on Weeks 1-3	Reproduced exactly on 12,000 rows	Continuity documented
Nested rolling-origin evidence; hold-out for confirmation	Test period already viewed in Week 3	7,200 predictions on later data	Stronger evidence
Cut-off NGN 142,058 to NGN 160,000	A scan picks it in every window	+0.006 ROC-AUC vs Week 3 features	Simpler, more stable
Keep Transaction_Status and Account_Status out	May not be known at decision time	+0.006 and -0.0003	Deployment-safe; cost documented
Burden-of-proof selection rule	Week 3 rule was borderline	No challenger qualified	Logistic Regression final
Default 20% review budget	The F1-optimal mode flags over half of all transactions	Budget table for 5-50%	Documented assumption
Keep six transaction types	Business-readable distinctions	S1 non-inferior (-0.0003)	Disclosed judgement call
Use the DA outputs for comparison only	Outputs were screenshots and code	74 of 74 values reconciled	Model unchanged

H. Assumptions, Limitations, Risks and Unresolved Issues

Assumptions

- Risk_Review_Flag is a consistent synthetic rule, not a fraud determination.
- The patterns in 1 Jan - 31 Mar 2026 persist for the transactions being scored.
- The NGN 160,000 cut-off, the night window and the 20% review budget are fixed inputs to be re-estimated if conditions change.

Limitations

- **Data ceiling:** ROC-AUC is about 0.67 and about two thirds of flagged transactions are false alarms at a 20% budget. A flexible model on every field does no better.
- **Short window:** three months, so no seasonality or drift can be assessed.
- **Test-set exposure:** the test period was viewed in Weeks 3 and 4; decisions rest on the nested evidence.
- **Scores:** few distinct values (ties); calibration was fitted on 2,400 rows.
- **DA outputs:** seen only as screenshots and code, so I reproduced each visible value rather than using their result tables.

Risks

- Reading the model as a fraud detector. It is a prioritization aid on synthetic data.
- Reusing the cut-off or budget on different data without re-estimating them.

Unresolved issues

- Whether Transaction_Status is available when a transaction is scored. If it is, it could add about 0.006 ROC-AUC.
- Integration with ML Engineering and GenAI has not happened.
- The DA suggestions have not yet been shared.

Lessons Learned

Three things that worked well

- Writing the decision rules before seeing the results made every feature and model choice defensible.
- Re-learning the amount cut-off and the hyper-parameters inside each rolling window showed that the amount step is real and stable.
- Reproducing the Week 3 dataset and candidate exactly gave a trustworthy starting point and a like-for-like comparison. Reconciling the DA dashboard values before using them did the same for the comparison.

Three challenges

- The data have a low ceiling: every model and feature set lands within about 0.02 ROC-AUC of the others.

- The test period had already been viewed in Week 3, so a second evaluation protocol had to be built.
- Reporting an honest result (the same ranking, a simpler and safer model) is harder than reporting a bigger number. The DA outputs also arrived as screenshots without result tables, so each value had to be reproduced.

Three lessons

- A simple, explainable model that is statistically equal to complex ones should win.
- A descriptive gap (channel, segment) is not the same as predictive value.
- A model is only finished when it behaves safely on bad input, not when its AUC has been computed.

Key improvements made. Four inputs instead of twelve (Week 2) and five (Week 3); a stable, learned amount cut-off; calibrated probabilities; a restated selection rule; nested evidence with intervals; a tested scoring function; and a reconciliation of the DA dashboard.

Recommendations for future work. Obtain a longer data window and real review-capacity figures; confirm whether transaction status is available at scoring time; reconcile KPIs with the DA dashboard as it evolves and share the night-time and amount-band suggestions; monitor drift in the amount cut-off; keep a human reviewer on every flag; and integrate with ML Engineering and GenAI once their outputs exist.

Week 4 Project Summary

1. **What I planned to accomplish in Week 4.** Test, validate, refine and finalise the predictive risk model, compare it with the Data Analytics findings, and document and present it.
2. **What I actually completed.** All Parts A-F: an exact reproduction of the Week 3 dataset and candidate, 20 feature variants, a nine-model comparison on nested and hold-out evaluation, error analysis, interpretation, a calibrated saved model with a model card, a validated scoring function, and a reconciliation of the DA dashboard (74 of 74 values).
3. **Tools used.** Python, Jupyter, pandas, NumPy, scikit-learn, SciPy, Matplotlib, Seaborn, joblib and Git/GitHub.
4. **Key findings.** Four signals matter: transaction type, night-time, international and a high-amount step at NGN 160,000. Nested ROC-AUC is 0.674 [0.658, 0.688]. At a 20% budget on the hold-out, precision is 0.357, recall 0.368 and lift 1.84x.
5. **Major challenges.** The data ceiling is low, the test period had already been viewed, and the honest result (the same ranking, a simpler and safer model) is less exciting than a bigger number.
6. **Important decisions.** Continue from the Week 3 dataset; nested evaluation as the main evidence; a pre-stated selection rule; keep status fields out; use the DA outputs for comparison only.
7. **Changes to my Week 3 approach.** Features cut from five to four, a learned amount cut-off, calibration added, and a safe scoring function.

8. **Testing and evaluation performed.** Nested rolling-origin evaluation, paired bootstrap, hold-out confirmation, shuffled-label, grouped cross-validation, threshold stability, subgroup checks, status sensitivity, reload test and 15 scoring tests (7 before, 15 after refinement).
9. **Key limitations.** Synthetic label, three-month window, modest ceiling, assumed budget, and DA outputs seen only as screenshots and code.